Real-Time Eye Tracking and Stress Analysis for Virtual Psychologists

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Introduction

- **The Context:** Teletherapy is expanding, but lacks the ability to read non-verbal cues.
- **The Problem:** Hidden signs of distress or anxiety often go undetected.
- **Our Solution:** A real-time, non-invasive eye-tracking system.
- **The Impact:** Monitors stress via gaze analysis, providing virtual therapists with insights into invisible emotional states.

Problems

- **The Sensory Gap** — Current virtual psychologists and teletherapy platforms are essentially 'blind' to non-verbal communication.
- **Invasive Hardware** — Traditional stress monitoring requires intrusive wearable sensors (ECG, EEG) or specialized laboratory equipment.
- **Privacy** — Recording raw video or audio for analysis raises significant privacy (GDPR) concerns.

Solution

Digital Sense

We empower virtual psychologists with a 'digital sense' through real-time eye tracking. This allows the system to detect non-verbal cues like gaze avoidance and blink rate instantly

Non-Invasive Analysis

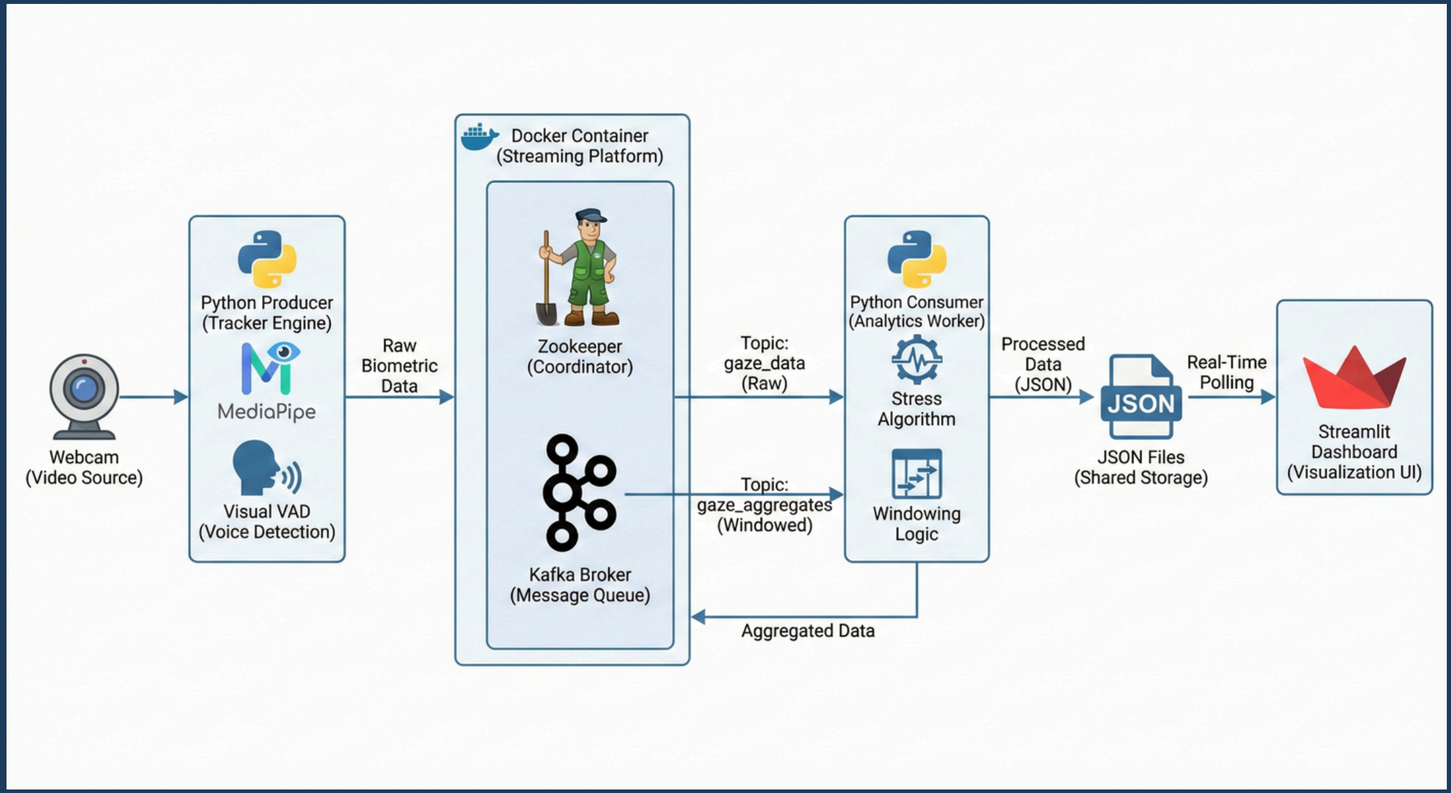
Our solution utilizes standard webcams and geometric algorithms, eliminating the need for expensive wearables. This makes advanced stress monitoring accessible and comfortable for any patient at home

Privacy-First Architecture

We prioritize patient confidentiality by processing only mathematical coordinates, not raw video or audio. Our Big Data pipeline ensures scalable, secure, and GDPR-compliant analysis

Methodologies & Technologies

1. Capturing and contextualizing biometric landmarks via Computer Vision (**MediaPipe**)
2. Extracting geometric metrics (EAR, MAR) for privacy-first behavioral analysis
3. Ingesting high-velocity real-time data streams and routing them using **Apache Kafka**
4. Processing stream data via rolling windows to calculate a composite Stress Score
5. Visualizing, analyzing, and monitoring real-time biometric feedback using **Streamlit**



Methodologies & Technologies

Eye Tracking Engine

Sub-Pixel Precision

478-landmark tracking via MediaPipe Face Mesh (Refined), including dedicated iris detection for sub-pixel precision.

Relative Vector Analysis

Relative iris-to-eye-corner vector replaces absolute pixel tracking, providing movement and distance invariance.

Dynamic Mesh

Dynamic eye mesh rendered via OpenCV, providing instant visual feedback on tracking quality and stability.

Voice Detection Engine

No Audio Recording

Full GDPR compliance: speech analysis based only on visual kinematics, with no audio capture or processing.

Geometric MAR Algorithm

MAR computed via normalize the distance between landmarks 13 & 14, quantifying mouth aperture independently of scale.

Dynamic Thresholding

The system triggers a 'Speaking' event only when the vertical mouth opening exceeds a calibrated threshold, filtering out noise and small movements

Kafka Implementation

Dual-Topic Strategy

Two dedicated data pipelines:

- gaze_data for real-time responsiveness,
- gaze_aggregate for windowed historical analysis.

The Envelope Pattern

Structured JSON wrapping decouples schema versioning from consumer pipelines, ensuring forward compatibility across system updates.

Windowing & Aggregation

Time-based windowing buffers incoming frames, computing stable statistical averages before storage via downsampling.

Stress Score Metric

4 Biometric Pillars

Blink Rate
Detects anxiety (high rate) or cognitive overload (low rate)

Gaze Variability
Measures restlessness via coordinate standard deviation.

Scan Velocity
Tracks the speed of saccadic eye movements.

Downward Gaze
Identifies avoidance behaviors or negative affect.

Reports about stress & eyetracking

Technical Challenges

- **Concurrency & I/O Locking**
  Continuous write/read operations on shared JSON files caused file-locking collisions in Windows, triggering access errors and pipeline failures.

- **Signal Jitter**
  Webcam sensor noise introduced micro-fluctuations in landmark coordinates, propagating into gaze vector instability and visual artifacts.

- **Schema Rigidity & Parallel Sessions**
  Initial flat data structure caused two problems: adding new metrics broke the system, and there was no way to separate data from different users.

Solutions

Exponential Backoff & Atomic Writes

Ensures data integrity via temporary files and automatic retry logic prevent data loss during collisions.

Temporal Smoothing Filter

Maintains stable tracking: moving average smoothing filters out signal jitter without user-perceptible delay.

Envelope Pattern & Kafka Key Partitioning

Guarantees scalable ordering: JSON envelopes separate metadata, while Session_id ensures ordered processing per user.

Future Implementation

LLM Integration

Feed the real-time Stress Score into a Large Language Model to adapt the Virtual Psychologist's empathy level dynamically.

Personalized Stress Detection

The system learns each patient's normal baseline in the first minutes, then detects personalized stress deviations.

Time-Series Database

Migrate from JSON files to InfluxDB for efficient long-term storage.

Demo

Thank you for your attention

Q&A time

Github Link